NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL 6117/1

PAPER 1 Multiple Choice 45 minutes

Marks 40 **2024**

Additional Materials: Multiple choice answer sheet
Non-programmable calculator
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Make sure that you receive the multiple choice answer sheet with **your examination number** on it.
- There are **forty** questions on this paper. Answer **all** questions.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.
- The Periodic Table is printed on page 11.
- **Read the instructions on the answer sheet carefully.**
- Each correct answer will score one mark.
- Any rough working should be done in this booklet.
- You may use a non-programmable calculator.

This document consists of **11** printed pages and **1** blank page.

Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

1 Which piece of apparatus is used to measure exactly 15.00 cm^3 of a liquid?

A 15 cm^3 beaker

B 20 cm^3 measuring cylinder

C 50 cm^3 conical flask

D 50 cm^3 burette

2 Some salt is contaminated with sand. How is a sample of solid salt obtained from the mixture?

A dissolve in water and then evaporate

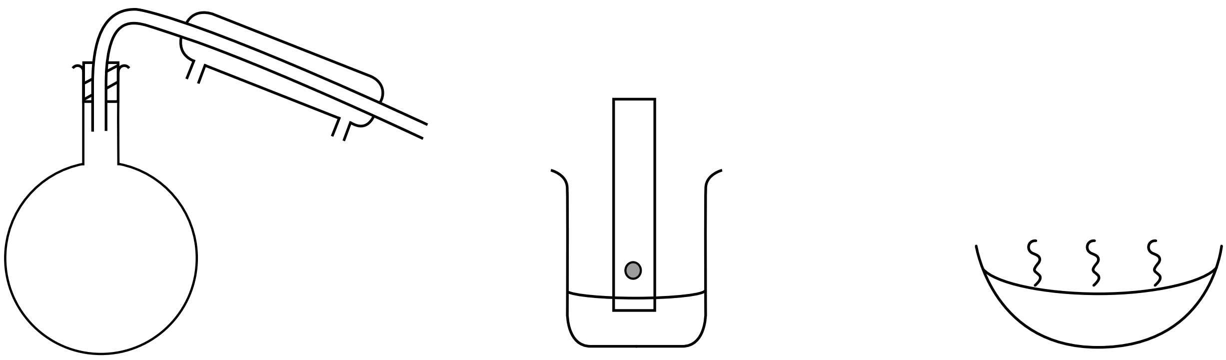
B dissolve in water, then filter and then dry the solid residue

C dissolve in water, then filter and evaporate the filtrate

D dissolve in water and then distil

3 The outline diagrams show three methods of purification.

P **Q** **S**



What are the three methods?

	P	Q	S
A	chromatography	distillation	evaporation
B	distillation	chromatography	evaporation
C	distillation	evaporation	chromatography
D	evaporation	chromatography	distillation

4 Which substance would diffuse most quickly?

A ammonia at 0°C

B ammonia at 25 °C

C carbon dioxide at 0 °C

D carbon dioxide at 25 °C

5 In which species is the numbers of electrons and neutrons equal?

A $^{9}_{4}Be$

B $^{19}_{9}F^{-}$

C $^{23}_{11}Na^{+}$

D $^{16}_{8}O^{2-}$

6 Which row about isotopes of atoms of the same element is true?

	same number of protons	same number of neutrons
A	✓	✓
B	×	×
C	✓	×
D	×	✓

7 Which property shows a decreasing trend in the elements of Group I, going down the group of the Periodic Table?

A density

B metallic character

C melting point

D reactivity with water

8 Which molecule contains one pair of shared electrons?

A ammonia

B chlorine

C nitrogen

D oxygen

9 Which statement is true for buckminsterfullerene?

A It is a nanomaterial.

B It is an isotope of carbon.

C It is heteronuclear.

D It consists of carbon and hydrogen.

10 How many different elements are present in ammonium hydroxide?

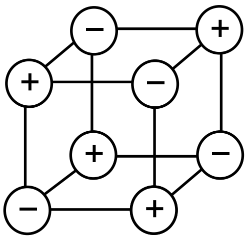
A 2

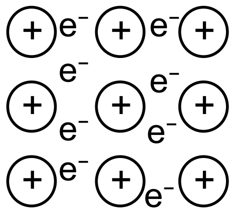
B 3

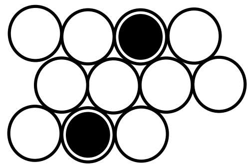
C 4

D 5

11 The diagrams show the arrangement of particles in three solids **X**, **Y** and **Z**.
The three solids are brass, potassium and sodium chloride.


X


Y


Z

Which row correctly identifies **X**, **Y** and **Z**?

	X	Y	Z
A	brass	potassium	sodium chloride
B	sodium chloride	brass	potassium
C	potassium	sodium chloride	brass
D	sodium chloride	potassium	brass

12 Which substance is used to hydrolyse fat during the preparation of soap?

A alkali

B oil

C dilute acid

D salt

13 Which term refers to an amount of substance containing approximately 6.022×10^{23} particles?

A Avogadro's constant

B a mole

C relative atomic mass

D relative formula mass

14 In a titration experiment, 25.0 cm^3 of 0.100 mol/dm^3 sodium hydroxide reacts with exactly 20.0 cm^3 of sulfuric acid according to the following equation.

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

What is the concentration of the sulfuric acid?

A 0.0625 mol/dm^3

B 0.0800 mol/dm^3

C 0.125 mol/dm^3

D 0.250 mol/dm^3

15 A compound contains 1.12 g of iron and 0.16 g of oxygen.

What is its empirical formula? [A_r: O = 16, Fe = 56]

A FeO

B Fe_2O

C FeO_2

D Fe_2O_2

16 Copper(II) sulfate can be electrolysed using carbon electrodes.

What forms at the cathode?

A copper

B hydrogen

C oxygen

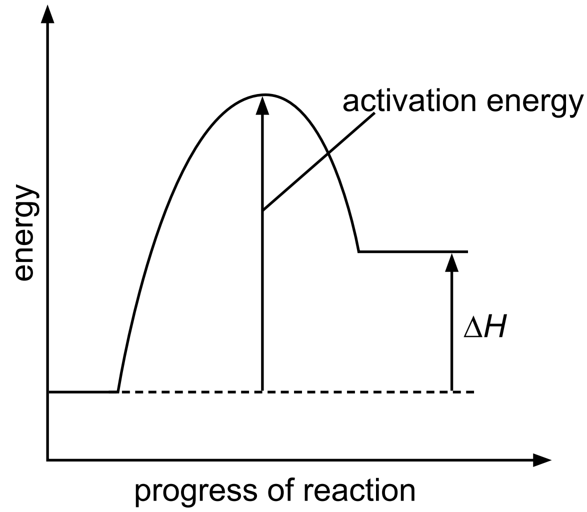
D sulfur

17 A spoon is to be electroplated with silver using an aqueous silver salt as the electrolyte.

Which row is correct?

	the object to be electroplated is the	**the other electrode is made from**
A	anode	carbon
B	anode	silver
C	cathode	carbon
D	cathode	silver

18 The energy level diagram for the forward direction of a reversible reaction is shown.



Which row correctly shows the sign of the activation energy and the type of enthalpy change?

	sign of the activation energy	**type of enthalpy change**
A	negative	endothermic
B	negative	exothermic
C	positive	endothermic
D	positive	exothermic

19 **X** is a substance that reacts with hydrogen in fuel cells.

What is **X**?

A carbon

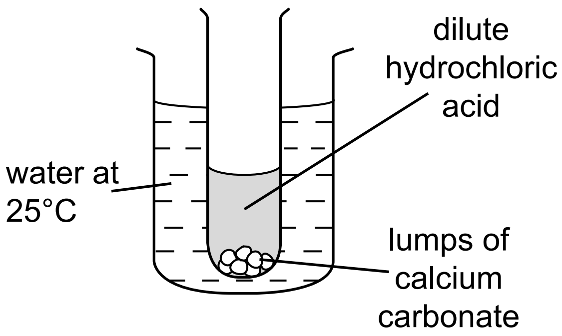
B methane

C nitrogen

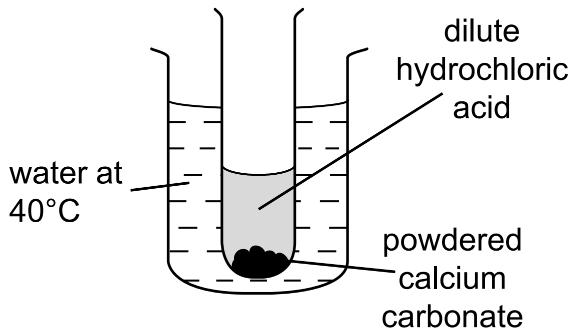
D oxygen

20 In which experiment is the rate of reaction between hydrochloric acid and calcium carbonate fastest?

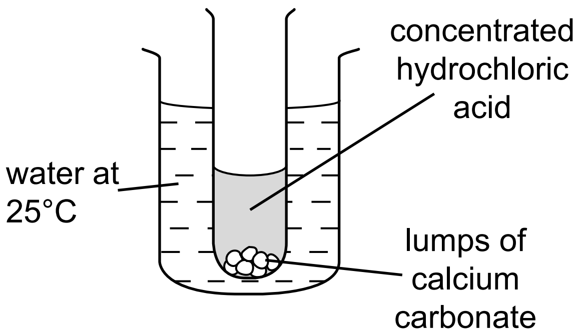
A



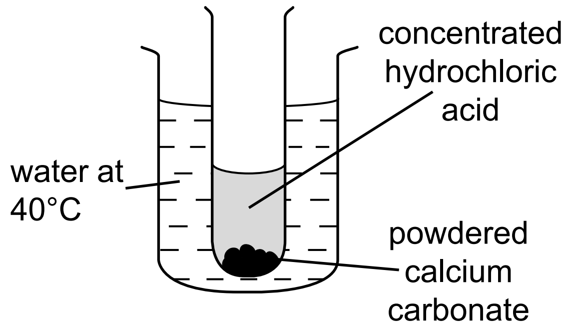
B



C



D



21 Which statement is true about enzymes?

A An enzyme is an organic catalyst that decreases the activation energy of a reaction.

B An enzyme is an organic catalyst that increases the activation energy of a reaction.

C An enzyme is an inorganic compound of a transition element that decreases the activation energy of a reaction.

D An enzyme is an inorganic compound of a transition element that increases the activation energy of a reaction

22 Which term refers to when the rates of the forward and backward reactions are equal?

A closed system

B dynamic equilibrium

C hydration

D reversible

23 Oxidation can be defined in terms of the gain or loss of oxygen and gain or loss of electrons.

Which row correctly describes the definitions of oxidation?

	oxygen	**electrons**
A	gain	gain
B	gain	loss
C	loss	loss
D	loss	gain

24 A learner tested some solutions with universal indicator paper. He wrote down their pH values 1, 5, 7 and 14 but forgot to write the names of the solutions.

Which solution did he correctly match with the pH?

	solution tested	**pH**
A	distilled water	1
B	concentrated sulfuric acid	5
C	sodium hydroxide	14
D	vinegar	7

25 Which combination of starting materials would produce copper(II) sulfate?

A Add copper to aqueous zinc sulfate.

B Add copper to dilute sulfuric acid.

C Add copper(II) carbonate to aqueous sodium sulfate.

D Add copper(II) oxide to dilute sulfuric acid.

26 Which metal gives a lilac colour when a flame test is carried out on it?

A barium

B calcium

C sodium

D potassium

27 What is used to test for the presence of hydrogen gas?

A damp red litmus paper

B glowing splint

C lighted splint

D limewater

28 Lead is more reactive than copper but less reactive than iron.

Which method would be most suitable for extracting lead from its ore?

A electrolysis

B heating alone

C heating with carbon

D reacting with hydrogen

29 Which statement about the reactions in the blast furnace is correct?

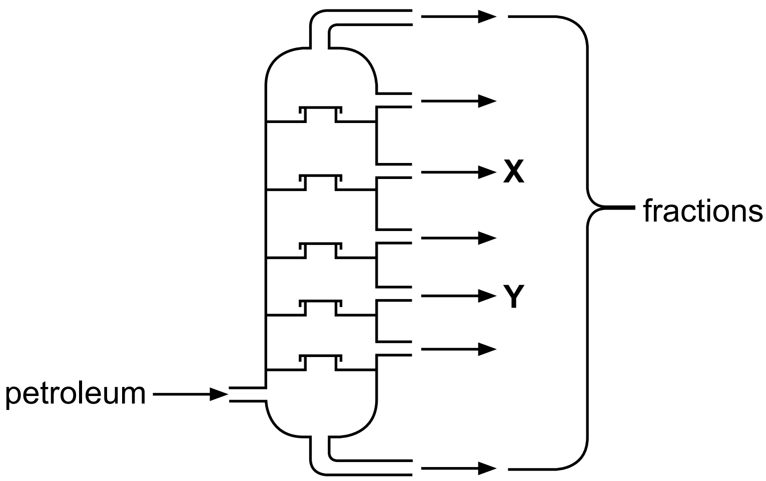
A Carbon reacts with oxygen to form carbon dioxide.

B Carbon monoxide removes the silicon dioxide impurity forming slag.

C Iron(III) oxide is oxidised to iron.

D Limestone reduces iron(III) oxide to iron.

30 Petroleum (crude oil) is separated into useful fractions by fractional distillation as shown in the diagram.



Which row shows the identity of **X** and **Y**?

	X	**Y**
A	diesel	naphtha
B	diesel	petrol
C	naphtha	petrol
D	naphtha	diesel

31 Which formula represents an alkane?

A $C_{11}H_{13}$

B $C_{11}H_{20}$

C $C_{11}H_{22}$

D $C_{11}H_{24}$

32 A learner does simple chemical tests to show that solution **Z** is unsaturated.

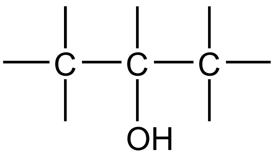
Which row shows the substance used for testing and the result obtained?

	substance	**result**
A	acidified potassium manganate(VII)	turns colourless
B	acidified potassium manganate(VII)	turns purple
C	litmus	bleached
D	litmus	turns blue

33 The diagrams show part of the structures of organic compounds.

Which compound can be manufactured by catalytic cracking?

A



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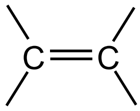

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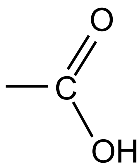

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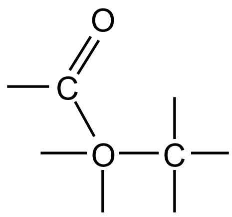
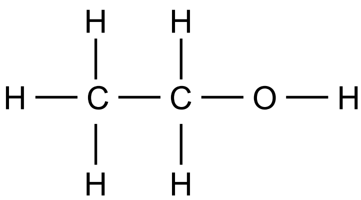


34 Which diagram shows the structure of ethanoic acid?

A



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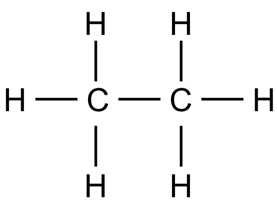

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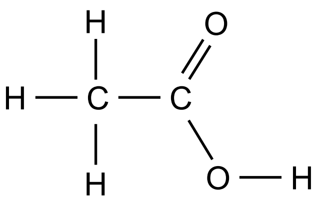

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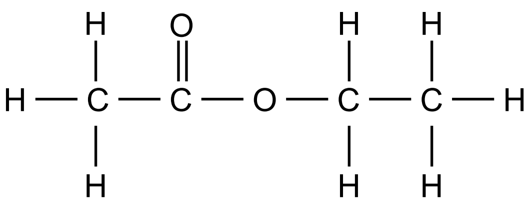
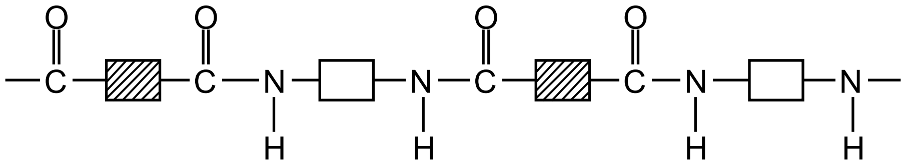


35 The diagram shows the partial structure of a polymer **S**.



What is **S**?

A fat

B nylon

C protein

D terylene

36 Which salt is dissolved in permanent hard water?

A $MgCl_2$

B $MgCO_3$

C $MgHCO_3$

D $MgSO_4$

37 Which process is used to kill bacteria during water treatment?

A chlorination

B eutrophication

C filtration

D sedimentation

38 Which substance gives off carbon dioxide on heating?

A lime

B limestone

C limewater

D slaked lime

39 The pH of a sample of garden soil is tested, and it is found to be 4.

What could be added to the soil to change its pH to 7?

A ammonium nitrate

B lime

C sand

D sodium chloride

40 Sulfuric acid is a strong acid.

Which of the following is a use of sulfuric acid?

A detergents

B bleach

C food preservative

D paper

DATA SHEET
The Periodic Table of the Elements

Group

I	II											III	IV	V	VI	VII	0
						1 **H** Hydrogen 1											4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4											11 **B** Boron 5	12 **C** Carbon 6	14 N Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12											27 **A*l*** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulfur 16	35,5 **C*l*** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 ***I*** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **T*l*** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	**Po** Polonium 84	**At** Astatine 85	**Rn** Radon 86
Fr Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89 †															

***58 - 71 Lanthanoid series**
†90 - 103 Actinoid series

140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60	**Pm** Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63	157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	162 **Dy** Dysprosium 66	165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69	173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71
232 **Th** Thorium 90	**Pa** Protactinium 91	238 **U** Uranium 92	**Np** Neptunium 93	**Pu** Plutonium 94	**Am** Americium 95	**Cm** Curium 96	**Bk** Berkelium 97	**Cf** Californium 98	**Es** Einsteinium 99	**Fm** Fermium 100	**Md** Mendelevium 101	**No** Nobelium 102	**Lr** Lawrencium 103

Key

a **X** b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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